

AVENGER FIELD, TX, USA

WMO: 722192

Lat: 32.473N

Lon: 100.466W

Elev: 2380

StdP: 13.48

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 23033

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.9	25.0	2.7	6.9	37.8	7.3	8.7	40.8	28.9	56.3	26.8	55.2	8.3	10	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.0	100.1	67.8	98.6	67.9	96.1	68.3	74.5	86.4	73.4	85.4	72.7	84.6	11.2	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.0	129.5	77.1	70.8	124.0	76.6	69.9	120.0	76.7	39.7	85.9	38.8	85.9	38.0	84.4	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
26.1	23.8	20.9	14.8	103.2	4.7	3.7	11.5	105.8	8.7	108.0	6.1	110.1	2.8	112.8		
			13.1	76.1	4.2	1.5	10.0	77.2	7.6	78.0	5.2	78.9	2.1	79.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	65.4	46.3	49.5	58.1	66.5	72.5	80.9	82.9	82.7	76.3	66.1	55.7	47.0
	DBStd	15.72	9.95	11.79	9.96	8.28	8.08	5.32	4.65	4.90	5.94	9.07	9.33	9.90
	HDD50	612	190	141	42	3	1	0	0	0	0	9	51	175
	HDD65	2347	582	444	253	81	28	0	0	0	4	99	297	559
	CDD50	6251	74	126	293	497	699	926	1020	1015	788	507	223	83
	CDD65	2508	1	9	39	125	261	476	555	550	342	131	18	1
	CDH74	28180	27	143	416	1362	2741	5618	6800	6547	3241	1107	154	24
	CDH80	13876	2	32	110	530	1261	2955	3675	3509	1425	353	23	1
Wind	WSAvg	10.8	11.2	11.5	12.1	13.2	11.8	11.5	9.3	8.5	8.7	10.3	11.0	10.6
Precipitation	PrecAvg	23.4	0.9	1.3	1.6	1.9	3.1	3.0	1.7	2.7	2.4	2.2	1.5	0.9
	PrecMax	34.1	2.7	4.8	4.8	3.9	7.4	7.4	4.6	7.7	6.7	5.7	4.7	5.7
	PrecMin	12.2	0.0	0.0	0.0	0.2	1.4	0.2	0.1	0.0	0.1	0.1	0.0	0.0
	PrecStd	6.3	0.7	1.3	1.1	1.3	1.7	2.2	1.3	2.3	1.9	1.6	1.3	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.4	85.1	88.3	94.8	99.5	103.7	102.3	102.4	98.6	91.0	82.8	77.4
		MCWB	52.4	55.5	57.9	59.8	64.6	67.9	67.8	68.6	67.1	64.4	58.7	54.0
	2%	DB	71.9	78.6	83.3	90.3	95.4	99.7	99.5	99.8	94.3	87.6	78.6	70.9
		MCWB	51.0	53.8	57.4	60.2	65.4	67.4	68.1	68.4	67.9	66.0	58.7	52.6
	5%	DB	66.6	73.0	79.1	86.4	91.1	96.9	97.3	97.6	91.2	84.2	74.2	66.5
		MCWB	49.7	52.8	56.8	60.5	65.3	68.5	68.9	68.8	67.3	65.3	57.1	51.6
	10%	DB	61.9	67.8	74.5	81.8	87.5	93.4	94.9	95.0	88.4	80.7	70.3	62.5
		MCWB	47.5	51.4	55.9	60.0	65.5	69.4	69.1	69.0	67.5	63.6	55.7	50.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.1	62.3	65.8	69.5	73.5	76.2	75.6	75.2	74.0	72.3	66.3	62.5
		MCDB	65.2	68.3	75.8	79.3	83.9	88.3	88.9	86.5	83.3	80.1	72.2	66.0
	2%	WB	55.3	60.0	63.8	67.9	72.1	74.5	74.1	73.8	72.7	70.5	63.6	58.7
		MCDB	64.5	68.9	72.8	77.8	82.8	86.9	86.9	85.8	81.4	79.8	70.1	63.8
	5%	WB	52.1	57.1	62.1	66.2	70.9	73.3	73.1	72.9	71.7	69.0	61.6	55.0
		MCDB	62.6	67.5	70.6	76.2	81.9	86.1	85.6	85.3	79.9	78.3	68.8	62.7
	10%	WB	49.0	53.4	59.9	64.2	69.4	72.3	72.2	72.0	70.8	67.2	59.1	51.1
		MCDB	60.2	65.9	69.5	73.7	80.9	85.1	84.7	84.6	79.5	75.8	67.4	61.8
Mean Daily Temperature Range	5% DB	MDBR	21.7	22.9	23.4	24.3	21.5	21.5	21.0	20.8	19.6	21.5	21.6	20.1
		MCDBR	27.5	30.3	28.5	29.1	26.8	25.9	24.0	23.9	23.1	24.3	24.9	26.5
		MCWBR	13.5	14.0	12.4	12.0	9.9	7.7	7.2	6.7	7.0	10.0	11.7	13.7
	5% WB	MCDBR	24.0	25.4	21.8	22.6	22.0	20.9	19.8	20.1	17.8	20.3	19.8	21.4
	MCWBR	14.5	15.4	11.5	11.4	9.8	7.7	6.8	6.7	6.1	10.6	12.8	14.3	
Clear-Sky Solar Irradiance	taub	0.279	0.291	0.315	0.350	0.368	0.395	0.402	0.391	0.363	0.341	0.308	0.288	
	taud	2.560	2.534	2.475	2.386	2.405	2.362	2.404	2.417	2.432	2.477	2.502	2.539	
	Ebn at Noon	303	309	306	298	292	282	279	282	286	286	288	293	
	Edh at Noon	26	29	33	38	38	39	37	36	35	30	27	25	
All-Sky Solar Radiation	RadAvg	1041	1270	1660	2030	2134	2306	2222	2046	1702	1374	1094	917	
	RadStd	109	160	113	143	200	153	142	126	121	188	125	97	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (17 neighbors)		+0.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+218	+149

Nomenclature: See separate page